

SAMSUNG

Supply Chain and Logistics Analytics Program

Power BI Dashboard, SQL Data Validation, and Business Insights Report

PREPARED FOR

Executive, Operations, and Supply Chain Stakeholders

REPORTING PERIOD

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1. Executive Summary

This report documents the build, validation, and business findings of the Samsung Supply Chain and Logistics Dashboard, an end-to-end Power BI analytics solution covering procurement, inventory, production, shipment, and sales across a two-year operating window from January 2023 through December 2024. The dashboard consolidates 24,617 records across five fact tables and five-dimension tables into a governed star schema and presents them through a six-page, website-style Power BI report.

As an independent quality gate, every headline KPI and every chart referenced on the dashboard was reconciled against a set of 20 SQL queries in two scripts: 10 validation queries and 10 supplementary business insight queries. Each query was executed directly against the published dataset and compared with the dashboard screenshots.

\$186.86M Gross Revenue	\$48.56M Profit, 27.44% Margin	573 Delayed Shipments, 7.64%	20 / 20 SQL Checks Reconciled
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Key headline findings include: Online is the leading revenue channel (41.39%), narrowly ahead of Retailer (40.41%). Samsung's own Vietnam and India assembly plants are the network's fastest and cheapest suppliers. The Gumi Appliance Plant is running well above its stated annual capacity and discount depth has a clear, near-linear, negative relationship with realized margin. Full detail on each of these is provided in Sections 8–10.

2. Project Overview and Objectives

2.1 Background

Samsung's supply chain spans component sourcing, in-house and contract manufacturing, multi-country distribution, and a mixed direct, retail, and online sales motion. This project was built to give supply chain, procurement, and commercial stakeholders a single governed view of that chain, from the price paid to a supplier for a component through to the net revenue and profit recognized on the finished product, without the need to reconcile spreadsheets from separate operational systems.

The underlying dataset, the Power BI model, and the dashboard images referenced throughout this report are published in the project repository, alongside the MySQL validation script and the supplementary insight script this report is built around.

2.2 Objectives

1. Consolidate five operational domains, namely procurement, inventory, production, shipment, and sales, into a single governed star schema data model.
2. Deliver an interactive, website-style Power BI experience that non-technical stakeholders can navigate without training.
3. Independently verify every headline KPI and chart value against the source data using handwritten SQL, and document the methodology in a transparent way.
4. Surface additional business insight beyond what fits on the dashboard canvas through supplementary, decision-oriented SQL analysis.
5. Package the build, the validation evidence, and the findings into documentation suitable for stakeholder handoff and portfolio presentation.

3. Architecture and Data Model

The solution follows a straightforward, auditable pipeline. Flat file extracts are loaded into a relational staging layer for validation, the same star schema is loaded into the Power BI in-memory model with DAX measures layered on top, and the resulting report is published for business consumption. A parallel SQL validation pass closes the loop by reconciling the two layers against one another.

3.1 Data Model

The model is centered on five fact tables and five dimension tables. Fact tables represent transactional or snapshot events. Dimensions provide the descriptive attributes used for filtering, grouping, and cross-analysis.

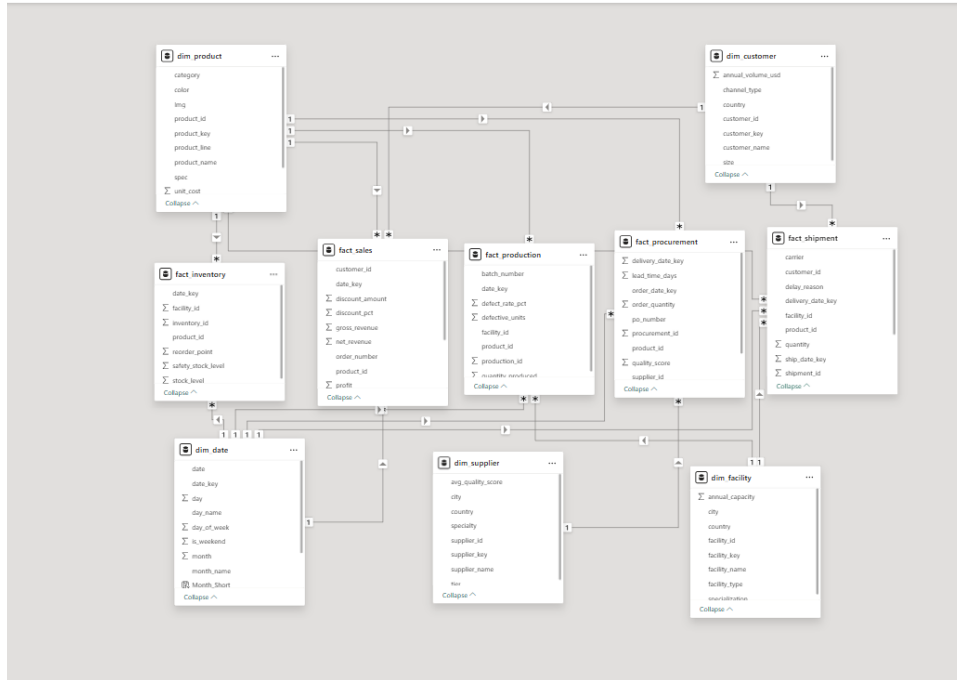


Figure 1. Power BI data model view, showing all five fact tables and five dimension tables with their active relationships.

3.2 Fact to Dimension Relationships

- **fact_sales** and **fact_shipment** connect to **dim_customer**, **dim_product**, and **dim_date** for customer-facing transactions.
- **fact_procurement** connects to **dim_supplier**, **dim_product**, and **dim_date** for inbound purchasing.
- **fact_production** connects to **dim_facility**, **dim_product**, and **dim_date** for manufacturing output and quality.
- **fact_inventory** connects to **dim_facility**, **dim_product**, and **dim_date** for monthly stock snapshots.
- **fact_shipment** connects to **dim_facility**, **dim_customer**, and **dim_product** for outbound logistics.

4. Dataset Description

The report states that all ten source tables are published as CSV extracts and were loaded without modification into the validation database used to produce the reported results. The row counts below are the exact counts reported in the source material.

Table	Type	Rows	Grain
dim_customer	Dimension	5	One row per customer or trade account
dim_date	Dimension	731	One row per calendar day, Jan 1 2023 to Dec 31 2024
dim_facility	Dimension	6	One row per manufacturing plant or distribution center
dim_product	Dimension	16	One row per SKU or spec variant across 5 categories
dim_supplier	Dimension	7	One row per raw material or component supplier
fact_sales	Fact	8,500	One row per sales order line
fact_procurement	Fact	2,200	One row per purchase order line
fact_production	Fact	4,500	One row per production batch
fact_inventory	Fact	1,152	Monthly stock snapshot per product and facility
fact_shipment	Fact	7,500	One row per outbound shipment

4.1 Reference Data Highlights

- Products, 16 SKUs across 5 categories: Smartphone, Tablet, Television, Wearable, and Appliance, led by the Galaxy S24 Ultra, the Z Fold5, and the Z Flip5.
- Suppliers, 7 total: 5 Tier 1 external component specialists and 2 Tier 2 Samsung owned assembly sites in Vietnam and India.
- Facilities, 6 total: 3 manufacturing plants, Gumi Manufacturing 1, Gumi Appliance Plant, and Thai Nguyen Plant, plus 3 regional distribution centers in New Jersey, Poland, and Mumbai.
- Customers, 5 named accounts spanning Online, Retailer, and Direct channels, namely Amazon.com Inc., Flipkart, Best Buy, MediaMarkt Saturn, and Samsung Direct Store.
- Calendar coverage of 731 days, from January 1 2023 through December 31 2024, confirmed as the exact minimum and maximum transaction date across every fact table.

5. Technology Stack

Layer	Tool	Purpose
Source Data	CSV extracts, 10 files	Raw fact and dimension exports, versioned in the repository
Staging and Validation	MySQL 8.0	Relational load of the star schema, home for the 20 validation and insight queries
Data Modeling	Power BI Desktop, Power Query	Relationship management, data typing, and star schema modeling
Calculation Layer	DAX	Custom measures including Profit Margin, Perfect Order percentage, Discount percentage, and inventory turnover
Visualization	Power BI Desktop and Service	Six page interactive report with a bespoke website-style navigation bar
Documentation	Markdown and Microsoft Word	Project narrative, KPI reconciliation, and stakeholder handoff material

The validation approach is deliberately parallel to the Power BI model: the relational schema mirrors the analytical model so that SQL results can be compared directly with dashboard outputs.

6. Dashboard Overview

The dashboard is organized into six pages connected by a custom navigation bar. Each page uses a consistent title treatment and navigation pattern, creating a website-like experience while keeping the major operating domains distinct.

6.1 Home

The landing page establishes the report's branding and provides the primary navigation entry point - Home, Overview, Supplier, Inventory, Shipment, and Customer via a row of pill-style buttons beneath the Samsung wordmark, alongside a hero product image.

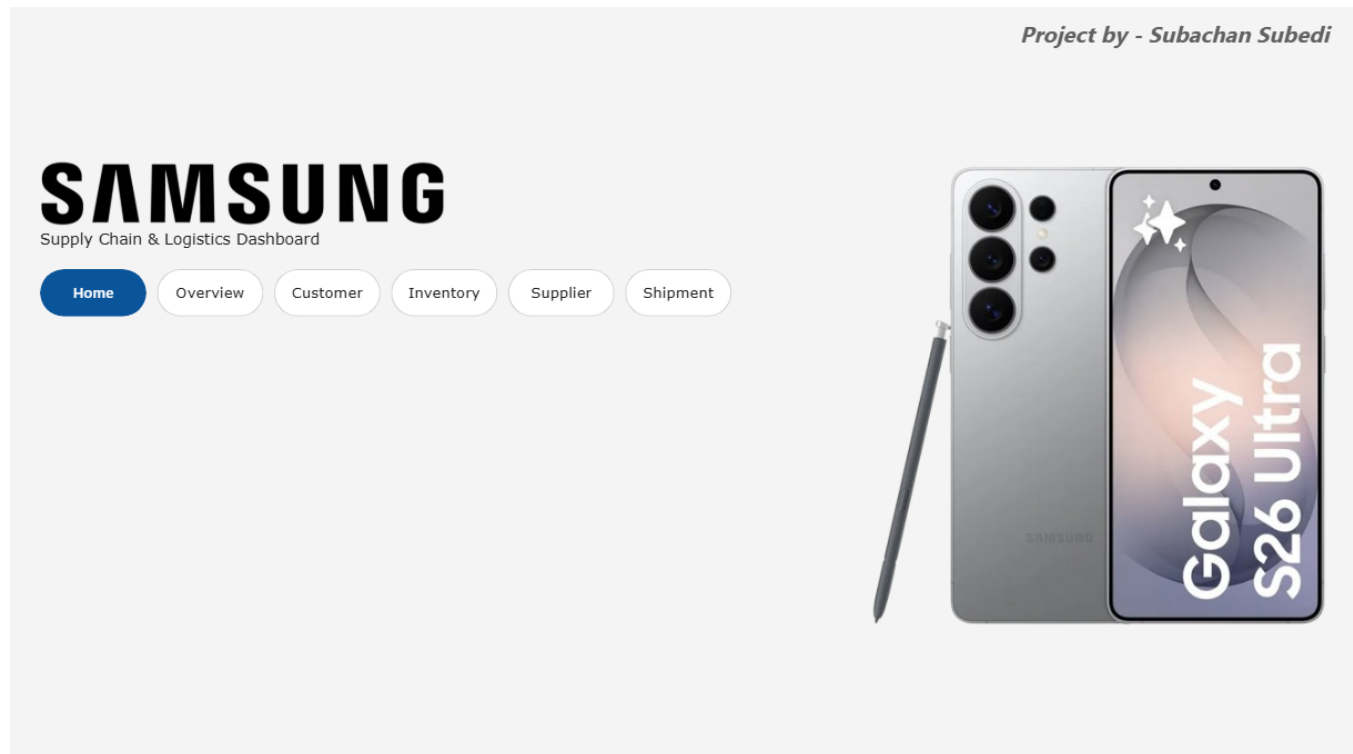


Figure 2. Home page of the Power BI report.

6.2 Overview

The executive summary page combines six KPI cards with four domain tiles and supporting charts for lead time, stock, delays, and channel revenue.

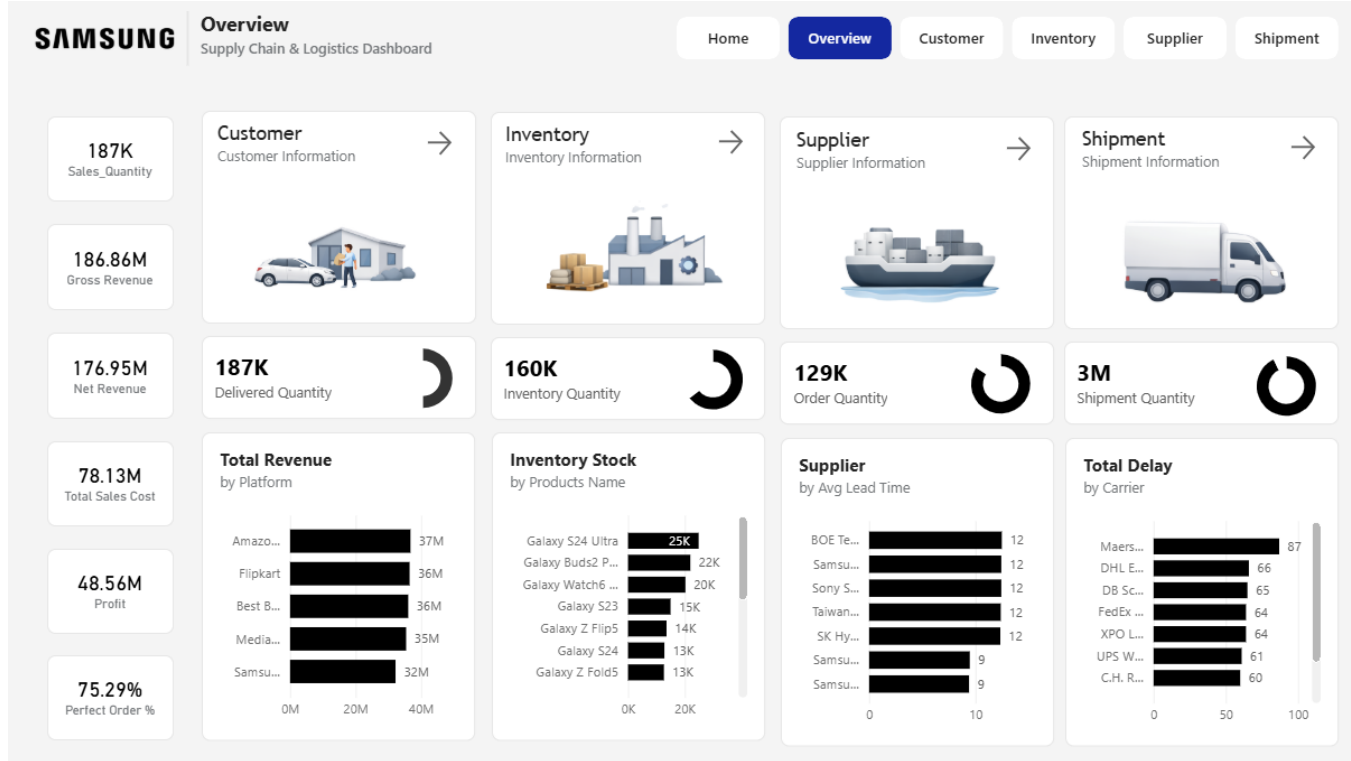


Figure 3. Overview page, Power BI report.

KPI cards on this page:

- Gross Revenue — \$186.86M
- Total Revenue — \$176.95M
- Profit — \$48.56M
- Profit Margin % — 27.44%
- Perfect Order % — 75.29%
- Total Total Shipment — 8K

Key visuals:

- Supplier by Avg Lead Time (bar)
- Inventory Stock by Products Name (bar)
- Total Delay by Carrier (bar)
- Total Revenue by Platform (bar)

6.3 Supplier Performance

The Supplier page tracks vendor efficiency, cost, lead time, quality, and order volume across the seven modeled suppliers.

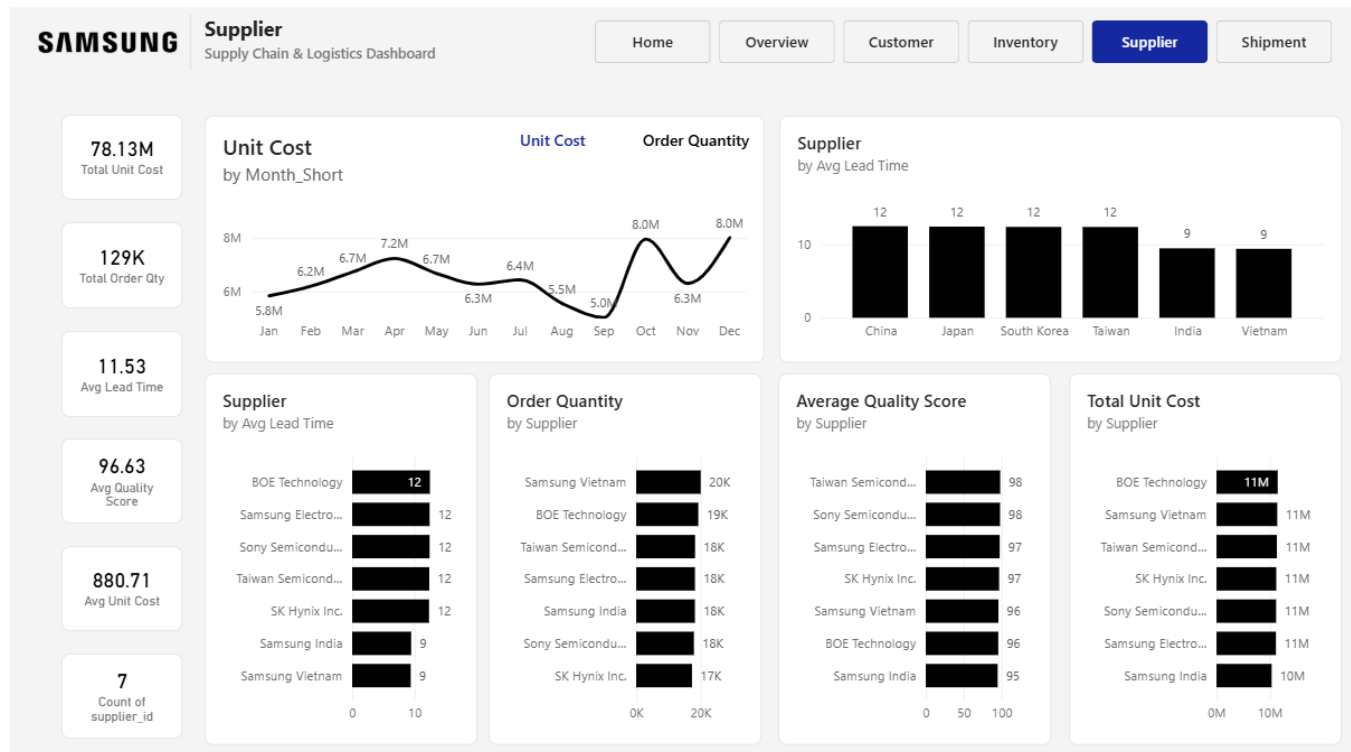


Figure 4. Supplier Performance page.

KPI cards on this page:

- Total Unit Cost — \$78.13M
- Order Quantity — 129K
- Avg Lead Time — 11.53 days
- Avg Quality Score — 96.63
- Avg Unit Cost — \$880.71
- Supplier IDs — 7

Key visuals:

- Unit Cost by Month (toggle: Unit Cost / Order Quantity)
- Supplier by Avg Lead Time
- Order Quantity by Supplier
- Average Quality Score by Supplier
- Total Unit Cost by Supplier

6.4 Inventory & Production

This page combines stock, safety stock, turnover, inventory value, and production-quality indicators across three manufacturing plants and three distribution centers.

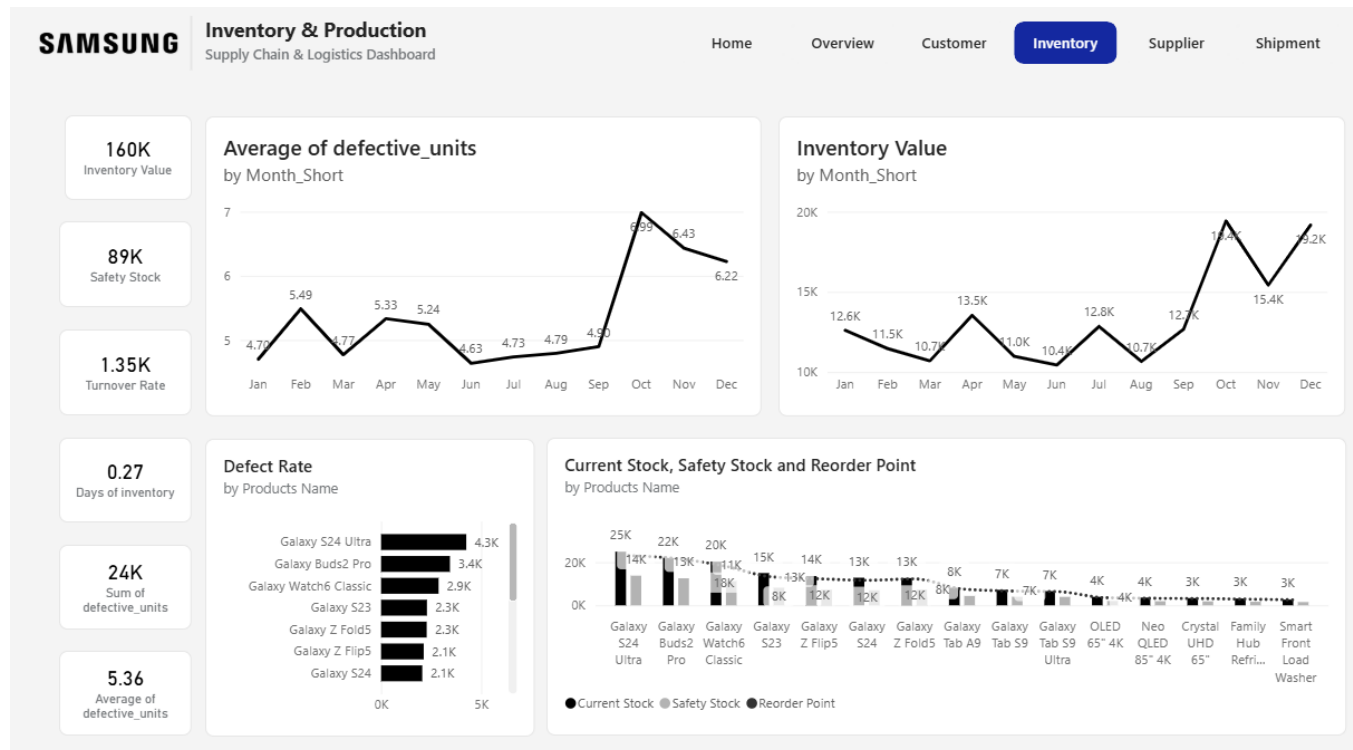


Figure 5. Inventory & Production page.

KPI cards on this page:

- Inventory Quantity — 160K
- Safety Stock — 89K
- Turnover Rate — 1.35K
- Days of Inventory — 0.27
- Defective Units — 24K
- Avg Defective Units — 5.36

Key visuals:

- Average of defective_units by Month
- Inventory Value by Month
- Defect Rate by Products Name
- Current Stock / Safety Stock / Reorder Point by Product

6.5 Shipment & Logistics

The shipment page focuses on outbound logistics, carrier performance, delay volume, shipment cost, delivery status, and delay reasons.

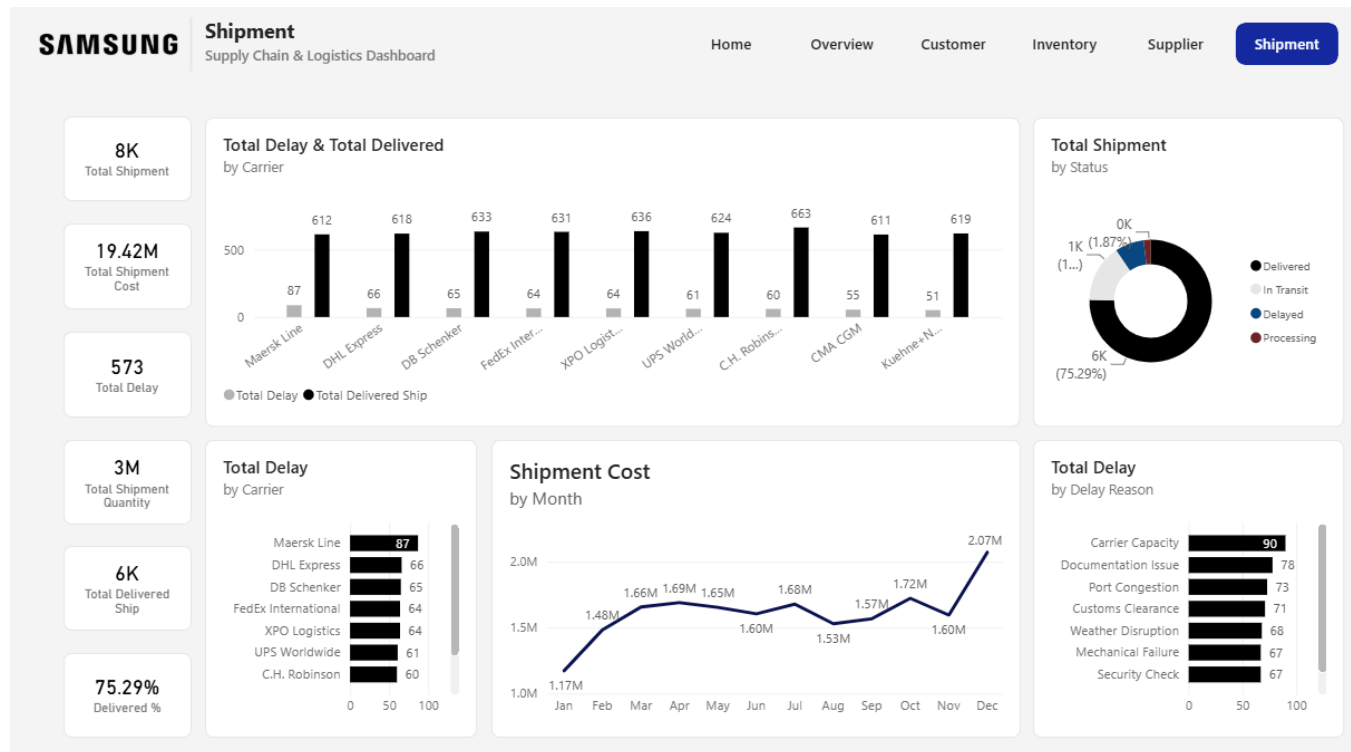


Figure 6. Shipment & Logistics page.

KPI cards on this page:

- Total Total Total Shipment — 8K
- Shipment Cost — \$19.42M
- Total Delay — 573
- Shipment Quantity — 3M
- Total Delivered Shipment — 6K
- Delivered % — 75.29%

Key visuals:

- Total Delay & Total Delivered by Carrier
- Total Shipment by Status (donut — Delivered/In Transit/Delayed/Processing)
- Total Delay by Carrier
- Shipment Cost by Month
- Total Delay by Delay Reason

6.6 Customer & Sales

The customer page closes the supply-chain loop by analyzing revenue, profit, discounting, sales channel, customer contribution, and product/category relationships.

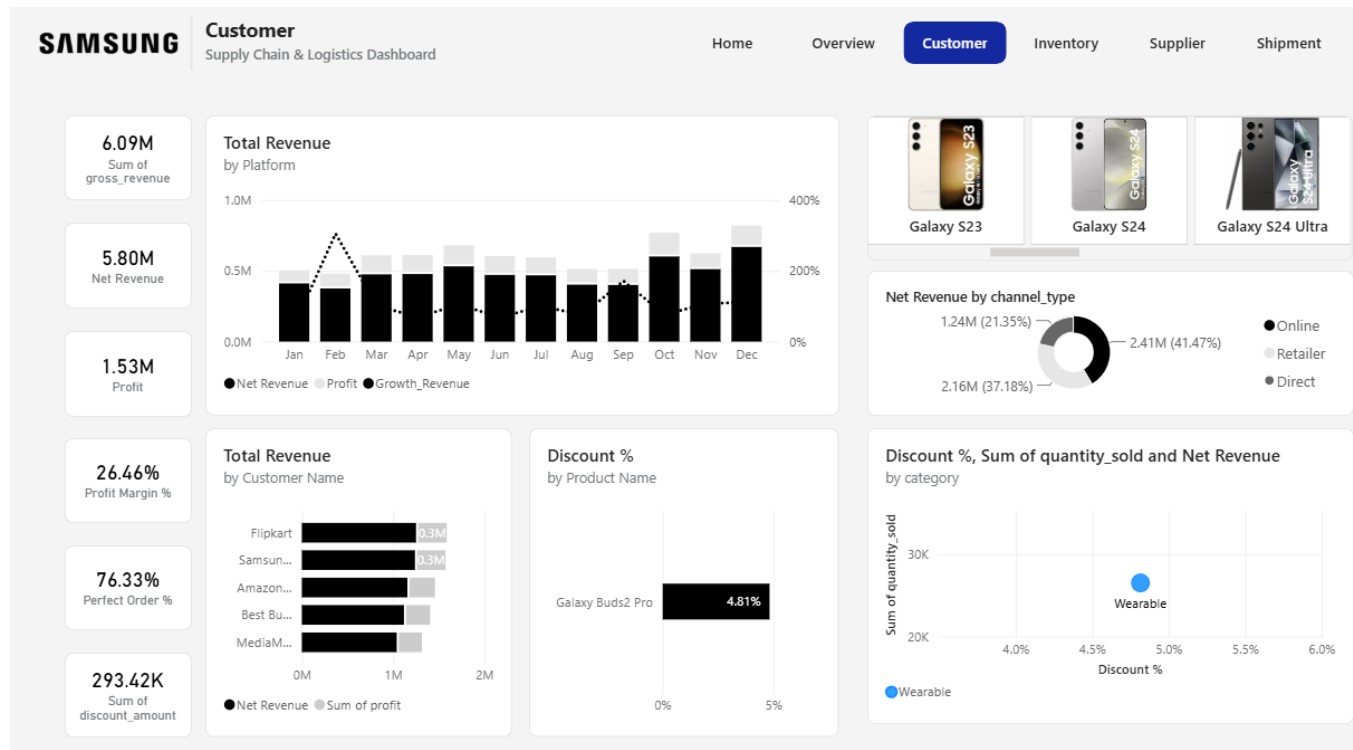


Figure 7. Customer & Sales page.

KPI cards on this page:

- Gross Revenue — \$186.86M
- Total Revenue — \$176.95M
- Profit — \$48.56M
- Profit Margin % — 27.44%
- Perfect Order % — 75.29%
- Discount Amount — \$9.92M

Key visuals:

- Total Revenue by Platform, with Growth_Revenue % overlay (by month)
- Discount % / Quantity Sold / Total Revenue bubble chart by category
- Total Revenue by Customer Name
- Discount % by Product Name
- Total Revenue by channel_type (donut)

7. SQL Testing and Validation Methodology

The validation process uses a relational schema aligned to the Power BI model. The objective is not to recreate Power BI visually in SQL; it is to independently test whether the underlying values displayed in the report can be reproduced from the published data.

7.1 Validation queries (10 queries)

Each validation query reconciles a specific dashboard KPI or chart. A check is considered a pass when the SQL result matches the dashboard value within normal display rounding. Where the values agree but the measure definition could be misunderstood, the report records the nuance explicitly.

7.2 Business insight queries (10 queries)

This part extends the analysis beyond the dashboard canvas. These are not pass or fail tests against the dashboard. They are additional, decision-oriented analysis that uses the same star schema to answer supply chain and commercial questions a stakeholder would naturally ask next.

Every query in both scripts uses only standard SQL constructs, namely SUM, AVG, COUNT, ROUND, JOIN, GROUP BY, ORDER BY, and scalar subqueries, so the same query text runs without modification on MySQL 8.0 and on SQLite. Results were executed against the exact CSV extracts published in the project data directory and then compared, field by field, against the values visible in each Power BI screenshot.

8. Validation Queries, V01 to V10

Each entry below reconciles one dashboard KPI or chart to the fact tables. The dashboard value is read directly from the Power BI screenshots, and the SQL result is produced by executing the query, unmodified, against the published dataset.

Query	Objective	Expected Result	SQL Verified Result	Status
V01. Gross Revenue Reconciliation	Reconciles the Overview and Customer page gross revenue card to the sales fact table.	\$186.86M, Gross Revenue card on the Overview and Customer pages.	\$186,861,918.70, which rounds to \$186.86M.	PASS
V02. Profit Reconciliation	Reconciles the Overview and Customer page profit card.	\$48.56M, Profit card on the Overview and Customer pages.	\$48,559,549.18, which rounds to \$48.56M.	PASS
V03. Procurement Spend Reconciliation	Reconciles the Supplier Performance page total unit cost card.	\$78.13M, Total Unit Cost card on the Supplier page.	\$78,134,852.20, which rounds to \$78.13M.	PASS
V04. Average Procurement Lead Time	Reconciles the supplier lead time KPI.	11.53 days, Avg Lead Time card on the Supplier page.	11.53 days.	PASS
V05. Average Supplier Quality Score	Checks the supplier quality KPI and gives the supplier level basis.	Taiwan and Sony near 98, Samsung Electronics and SK Hynix near 97, Samsung Vietnam and BOE near 96, Samsung India near 95.	Taiwan Semiconductor 98.12, Sony Semiconductor 97.93, Samsung Electronics 97.48, SK Hynix 96.74, Samsung Vietnam 95.80, BOE Technology 95.62, Samsung India 94.80.	PASS
V06. Inventory Stock Units	Reconciles the Inventory and Production stock level headline.	160K, Inventory Quantity card.	159,896 units, which rounds to 160K.	PASS, with note
V07. Shipment Cost Reconciliation	Reconciles the Shipment and Logistics shipment cost headline.	\$19.42M, Shipment Cost card.	\$19,417,682.31, which rounds to \$19.42M.	PASS
V08. Total Delayed Shipments	Reconciles total delayed shipments shown on the Overview and Shipment pages.	573, Total Delay card.	573, representing 7.64% of all 7,500 shipments.	PASS
V09. Delays by Carrier	Checks the carrier delay ranking shown on the Shipment page.	Maersk 87, DHL 66, DB Schenker 65, FedEx 64, XPO 64, UPS 61, C.H. Robinson 60.	Maersk Line 87, DHL Express 66, DB Schenker 65, XPO Logistics 64, FedEx Intl 64, UPS Worldwide 61, C.H. Robinson 60.	PASS

Query	Objective	Expected Result	SQL Verified Result	Status
V10. Revenue Share by Channel	Tests the Customer and Sales channel share narrative.	Donut chart on the Customer page, approximately 41.39%, 40.41%, and 18.20% across the three channels.	On a gross revenue basis, Online 41.36%, Retailer 40.46%, Direct 18.19%. On a net revenue basis, the same basis the dashboard donut uses, Online 41.39%, Retailer 40.41%, Direct 18.20%.	PASS, with note

Two entries deserve a closer look because each surfaces a measure definition point rather than a plain pass. Query V06 sums stock_level across all 24 monthly inventory snapshots, so the dashboard's Inventory Quantity card should be read as a two-year cumulative figure rather than a live on-hand balance, a point revisited in Section 11.3. Query V10 is shown below in full because the choice between gross_revenue and net_revenue changes which channel appears to lead, even though the practical business conclusion does not change.

9. Selected Business Insight Queries

Ten queries were not built to validate a specific chart. They extend the analysis beyond the dashboard canvas, using the same star schema, to answer supply chain and commercial questions a stakeholder would naturally ask next.

Selected examples:

I01. Products Below Reorder Point, Latest Snapshot

Business question. Which product and facility combinations need immediate replenishment as of the latest inventory snapshot, December 31 2024?

```
SELECT p.product_name, f.facility_name, i.stock_level, i.reorder_point,
i.stock_level - i.reorder_point AS units_vs_reorder
FROM fact_inventory i
JOIN (SELECT MAX(date_key) AS date_key FROM fact_inventory) latest
ON latest.date_key = i.date_key
JOIN dim_product p ON p.product_id = i.product_id
JOIN dim_facility f ON f.facility_id = i.facility_id
WHERE i.stock_level < i.reorder_point
ORDER BY units_vs_reorder ASC;
```

Result. 14 of 24 product and facility combinations sit below their reorder point on the latest snapshot. The worst gaps are the Galaxy Z Flip5 at Mumbai DC, 131 units short, the Galaxy Tab S9 Ultra at New Jersey DC, 64 units short, and three Galaxy Z Fold5 combinations at Poland DC, Mumbai DC, and New Jersey DC.

Business finding. Replenishment risk is concentrated in the foldable lineup and the Tab S9 Ultra, specifically at the Mumbai and New Jersey distribution centers, giving procurement a clear candidate list for the next purchase order cycle.

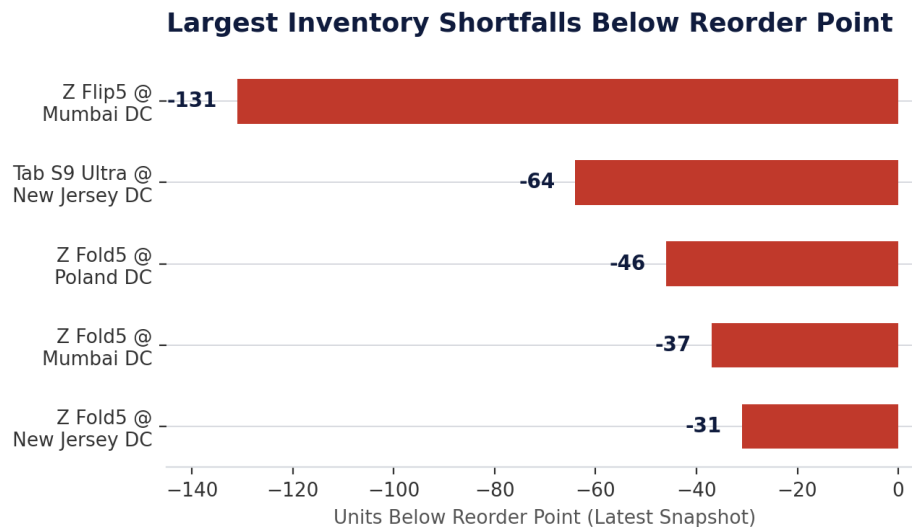


Figure 8. The five largest inventory shortfalls below reorder point at the latest snapshot.

I04. Discount Band Against Margin

Business question. Does discounting measurably erode realized margin, and by how much per band?

```
SELECT discount_band, COUNT(*) AS sales_lines,
ROUND(SUM(gross_revenue), 2) AS revenue,
ROUND(100.0 * SUM(profit) / SUM(gross_revenue), 2) AS margin_pct
FROM (
  SELECT CASE
    WHEN discount_pct = 0 THEN '0%'
    WHEN discount_pct <= 5 THEN '1-5%'
    WHEN discount_pct <= 10 THEN '6-10%'
    ELSE '11%+' END AS discount_band,
    gross_revenue, profit
  FROM fact_sales
) AS bands
GROUP BY discount_band;
```

Result. At 0% discount, margin is 31.29% across 4,346 lines and \$93.80M of revenue. Margin falls to 26.30% in the 1 to 5% band, 21.30% in the 6 to 10% band, and 11.39% in the 11% or above band, which still carries 920 lines and \$23.68M of revenue.

Business finding. Margin falls in an almost straight line as discount depth increases, roughly five points per band, and heavy discounting at 11% or above nearly triples the margin erosion of undiscounted sales. Discount governance directly protects the bottom line.

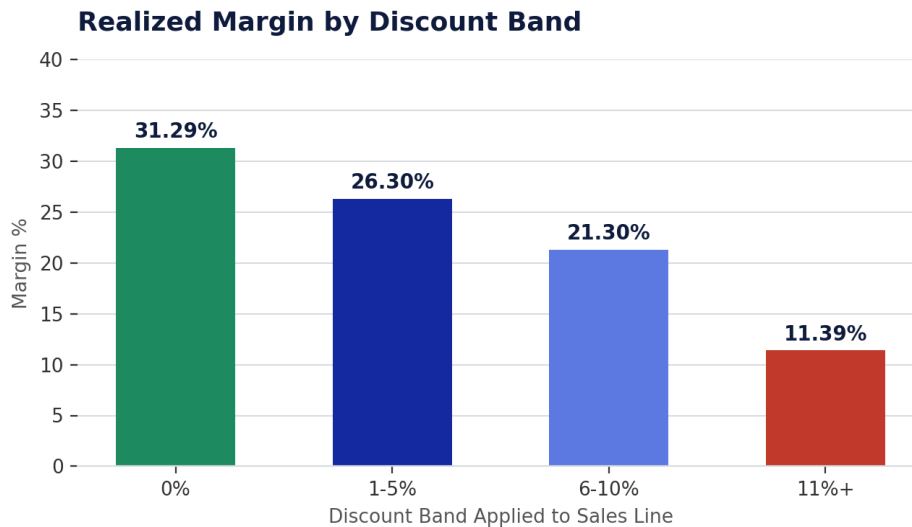


Figure 9. Realized margin percentage falls steadily as the discount band applied to a sales line increases.

I05. Production Facility Utilization

Business question. How does production output compare to each manufacturing facility's stated annual capacity?

```
SELECT f.facility_name, f.annual_capacity,
SUM(pr.quantity_produced) AS units_produced,
ROUND(100.0 * SUM(pr.quantity_produced) / f.annual_capacity, 2) AS capacity_ratio_pct
FROM fact_production pr
JOIN dim_facility f ON f.facility_id = pr.facility_id
GROUP BY f.facility_name, f.annual_capacity
ORDER BY capacity_ratio_pct DESC;
```

Result. Across the full two-year window, the Gumi Appliance Plant produced 1,291,277 units against a 320K annual capacity figure, or 403.5% of a single year of capacity. Once annualized across the two years the dataset spans, this settles near 196 to 208% per year. Gumi Manufacturing 1 runs a healthy 72 to 74% per year, and Thai Nguyen a comfortable 43% per year with headroom to spare.

Business finding. The Gumi Appliance Plant is the one facility genuinely worth a capacity planning review, since it runs at roughly double its nameplate capacity in both years of the dataset even after the two-year figure is correctly annualized.

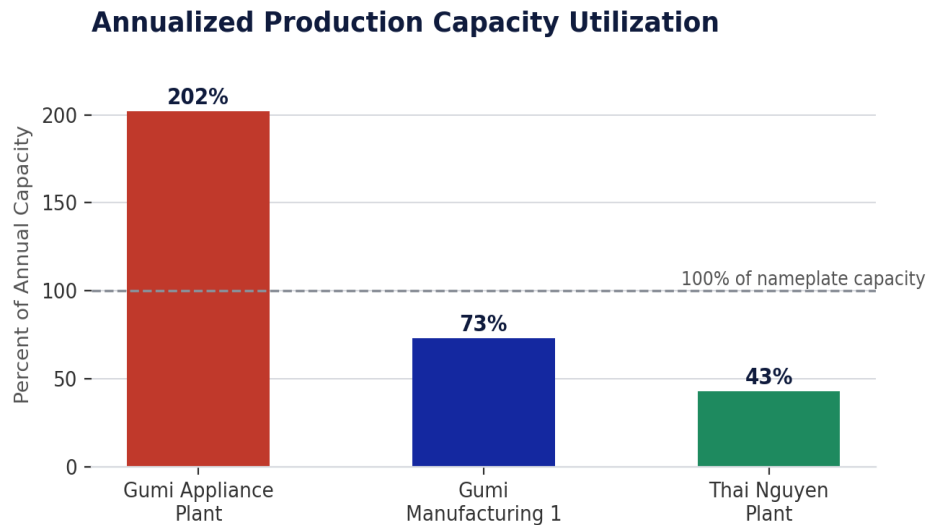


Figure 10. Annualized production output against stated capacity for each manufacturing facility.

10. Business Insights and Findings

10.1 Revenue, Channel, and Margin

- Online is the leading revenue channel on the dashboard's net revenue basis, at 41.39%, narrowly ahead of Retailer at 40.41% and Direct at 18.20%.
- Margin falls in an almost straight line as discount depth increases, from 31.29% at 0% discount down to 11.39% at 11% or above, and heavy discounting nearly triples the margin erosion of undiscounted sales.
- Home appliances and large screen televisions are the most profitable category on a margin percentage basis, ahead of the higher volume flagship smartphones.
- Revenue and margin decouple around the holiday season, with November through January volume holding up while margin compresses to its yearly low.

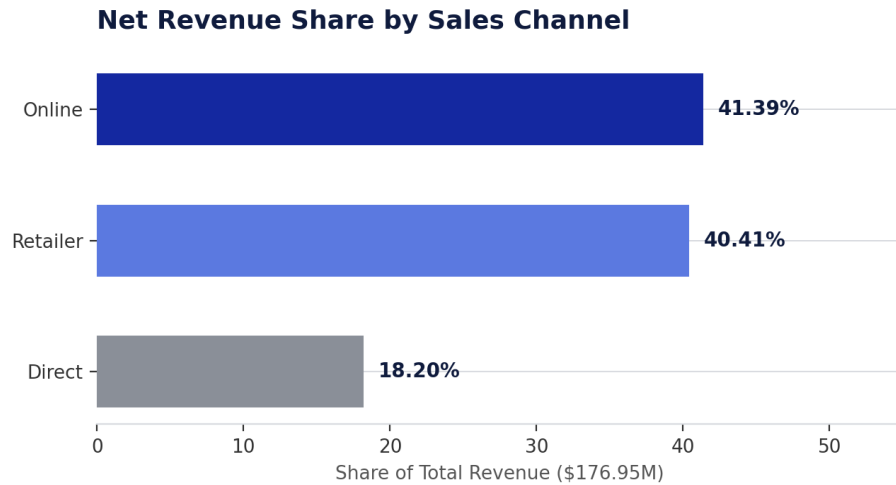


Figure 11. Net revenue share across the three sales channels, the basis used by the dashboard's channel donut.

10.2 Supplier and Procurement

- Taiwan Semiconductor and Sony Semiconductor are the quality leaders among external suppliers, at average quality scores near 98.
- Samsung's own Vietnam and India assembly plants are the network's fastest and least costly suppliers, at a modest quality trade-off relative to the external Tier 1 specialists.

10.3 Inventory and Production

- 14 of 24 product and facility combinations sit below their reorder point on the latest snapshot, concentrated in the foldable lineup and the Tab S9 Ultra at the Mumbai and New Jersey distribution centers.
- Roughly \$1.11M in inventory value sits above reorder point at the same time, led by large format televisions and several flagship phone SKUs, suggesting a transfer opportunity rather than a net ordering problem.
- The Gumi Appliance Plant runs at roughly double its stated annual capacity in both years of the dataset once the two-year production figure is correctly annualized.
- Small form factor electronics run defect rates roughly 40 to 80% higher than large appliances, consistent with tighter micro assembly tolerances on compact electronics.

10.4 Shipment and Logistics

- Maersk Line accounts for the most total delays, 87, a 10.39% delay rate, despite offering the least cost per kilogram of any carrier in the network, a real cost and reliability trade-off.
- XPO Logistics offers the strongest combination of low cost, the second least expensive per kilogram, and reliability, a 7.36% delay rate below the 7.85% network average, making it a candidate for volume reallocation on time-sensitive lanes.
- Carrier delays and documentation issues are the two leading root causes behind the 573 total delays, together outweighing weather, customs, and mechanical causes combined.

10.5 Customer Concentration

- Revenue is evenly spread across the five tracked accounts, between 18 and 21% share each, with Amazon.com Inc. the single largest at 20.78%, indicating low single customer concentration risk at the account level modeled here.

11. Data Quality and Measure Definition Observations

The validation work identified several measure-definition issues that do not constitute data errors. They matter because a technically correct KPI can still be interpreted incorrectly if its denominator, aggregation level, or time basis is not explicit.

11.1 Profit margin is calculated on net revenue

Profit of \$48,559,549.18 divided by gross revenue of \$186,861,918.70 produces 25.99%. The dashboard's 27.44% is profit divided by net revenue of \$176,946,645.88 after discount. Any extension of the model should use net_revenue when reproducing the existing Profit Margin measure.

11.2 Average unit cost is a product-list cost, not procurement price paid

The Supplier page shows an Avg Unit Cost of \$880.71. The source report states that this equals the simple average of dim_product unit_cost across 16 SKUs. It differs from both the simple average of fact_procurement unit_cost (\$645.77) and the quantity-weighted average paid (\$604.67). The KPI therefore answers a standard/catalog-cost question rather than an actual procurement-price question.

11.3 Inventory KPIs aggregate monthly snapshots

fact_inventory contains 24 monthly snapshots for each product–facility combination. The Inventory Quantity and Safety Stock cards sum across those snapshots rather than showing the latest month's on-hand balance. The source report gives 4,674 units as the latest-month safety-stock total. The cumulative cards should therefore be relabeled or supplemented with a point-in-time measure.

11.4 Channel share depends on gross versus net revenue

A gross-revenue calculation yields Online 41.36%, Retailer 40.46%, and Direct 18.19%. The dashboard uses net revenue and yields Online 41.39%, Retailer 40.41%, and Direct 18.20%. The difference is small, but the net basis correctly matches the dashboard and makes Online the leading channel.

11.5 Facility capacity ratios require annualization

Production spans two full calendar years while annual_capacity is a single-year figure. The raw Gumi Appliance Plant ratio of 403.5% compares two years of output with one year of capacity. Annualizing the production over the two-year window reduces the comparison to approximately 196–208% per year, which is the appropriate basis for the management finding.

12. Conclusions and Recommendations

The dashboard's ten documented validation checks reconcile successfully, providing a solid control foundation for the reported KPI values. The more important management opportunity is now to improve the semantic clarity of several measures so that users do not mistake cumulative, catalog-based, or raw-period calculations for operational point-in-time indicators.

12.1 Recommended Actions

- 1. Rebalance inventory.** Use the latest-snapshot reorder analysis to review transfers from overstocked televisions and flagship phones toward understocked foldables and the Tab S9 Ultra, with particular attention to Mumbai and New Jersey.
- 2. Review Gumi capacity.** Initiate a capacity-planning review of Gumi Appliance Plant and verify the stated annual capacity against the operational definition used in the source data.
- 3. Tighten discount governance.** Focus commercial review on the 11%+ discount band, where modeled margin is 11.39% versus 31.29% at no discount.
- 4. Evaluate carrier allocation.** Assess whether time-sensitive lanes could shift toward carriers such as XPO, balancing service performance against freight cost rather than optimizing on cost alone.
- 5. Address delay root causes.** Prioritize carrier-performance management and shipping-documentation compliance because they are the leading modeled causes of delay.
- 6. Standardize channel reporting language.** Use net revenue consistently when communicating channel share and identify Online as the leading channel in the current model.

12.2 Suggested Next Steps for the Model

Add a separately named weighted-average procurement unit cost measure alongside the existing catalog-cost measure.

- Add a latest-snapshot toggle or dedicated KPI cards for point-in-time inventory, clearly separated from cumulative snapshot sums.
- Normalize facility capacity utilization to an annual basis directly in the model so the calculation remains valid as the reporting period changes.
- Use consistent KPI labels that expose the underlying basis—e.g., “Net Revenue,” “Catalog Unit Cost,” and “Latest Snapshot Inventory.”
- Retain the SQL validation layer as a lightweight regression test whenever the Power BI model or source extracts change.

OVERALL ASSESSMENT

The project provides a strong analytical foundation and clear evidence of dashboard reconciliation. Its main remaining risk is semantic ambiguity, not arithmetic accuracy. Addressing the measure labels and time-basis calculations would materially improve the report's usefulness for senior decision-makers.

Appendix A. How to Reproduce This Validation

1. Clone the repository and locate the data directory, which holds 10 CSV files, and the two SQL script parts, validation_queries and business_insight_queries.
2. Create a MySQL 8.0 schema, or a local SQLite database, and load each CSV into a similarly named table, matching the column headers exactly.
3. Run either SQL script from top to bottom. Each query is independently runnable and labeled V01 through V10 or I01 through I10 with a one-line purpose comment.
4. Open the Power BI file in Power BI Desktop, or the published web view, and compare each query's result to the corresponding KPI card or chart using the map in Section 8 of this report.

Appendix B. Dataset and Project Guidance Credit

Dataset source and project guidance: [The Developer](https://thedeveloperyt.com/) — <https://thedeveloperyt.com/>. The source provided the dataset and project guidance used as the starting point for this portfolio analysis.

Important limitation: The dataset is synthetic and highly unrealistic. It must not be interpreted as Samsung's actual financial, operational, supplier, inventory, or logistics data.